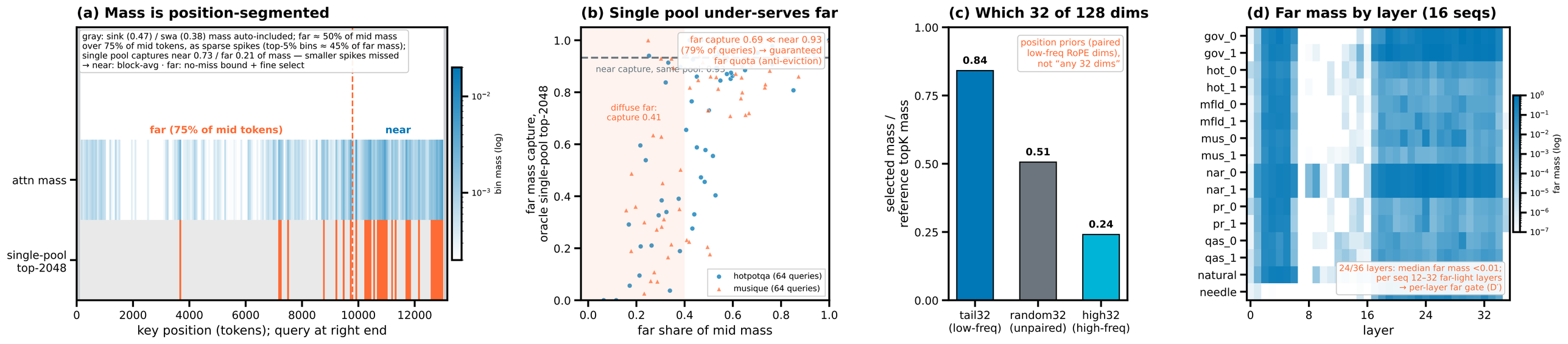

(a) Mass is position-segmented
gray: sink (0.47) / swa (0.38) mass auto-included; far ≈ 50% of mid mass over 75% of mid tokens, as sparse spikes (top-5% bins ≈ 45% of far mass); single pool captures near 0.73 / far 0.21 of mass — smaller spikes missed → near: block-avg · far: no-miss bound + fine select
far (75% of mid tokens)
near
attn mass
single-pool top-2048
bin mass (log)
key position (tokens); query at right end
(b) Single pool under-serves far
far capture 0.69 ≪ near 0.93 (79% of queries) → guaranteed far quota (anti-eviction)
near capture, same pool: 0.93
diffuse far: capture 0.41
hotpotqa (64 queries)
musique (64 queries)
far mass capture, oracle single-pool top-2048
far share of mid mass
(c) Which 32 of 128 dims
position priors (paired low-freq RoPE dims), not “any 32 dims”
0.84
0.51
0.24
selected mass / reference topK mass
tail32 (low-freq)
random32 (unpaired)
high32 (high-freq)
(d) Far mass by layer (16 seqs)
gov_0
gov_1
hot_0
hot_1
mfld_0
mfld_1
mus_0
mus_1
nar_0
nar_1
pr_0
pr_1
qas_0
qas_1
natural
needle
24/36 layers: median far mass <0.01; per seq 12–32 far-light layers → per-layer far gate (D′)
layer
far mass (log)